

shows that the only singularities $\text{sn}(z)$ are poles. Functions of this type, called “elliptic functions,” are the subject of the next chapter.

5 Exercises

1. A holomorphic mapping $f : U \rightarrow V$ is a **local bijection** on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z , so that $f : D \rightarrow f(D)$ is a bijection.

Prove that a holomorphic map $f : U \rightarrow V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché’s theorem as in the proof of Proposition 1.1.]

2. Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0 , are orthogonal at z_0 , and so that F restricted to Γ_1 is real and has a minimum at z_0 , while F restricted to Γ_2 is also real but has a maximum at z_0 .

[Hint: Write $F(z) = (g(z))^2$ for z near z_0 , and consider the mapping $z \mapsto g(z)$ and its inverse.]

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V . Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V .

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

[Hint: Move the upper half-plane “down” and then square it to get $\mathbb{C}$.]

5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

6. Give another proof of Lemma 1.3 by showing directly that the Laplacian of $u \circ F$ is zero.

[Hint: The real and imaginary parts of F satisfy the Cauchy-Riemann equations.]

7. Provide all the details in the proof of the formula for the solution of the Dirichlet problem in a strip discussed in Section 1.3. Recall that it suffices to compute the solution at the points $z = iy$ with $0 < y < 1$.

(a) Show that if $re^{i\theta} = G(iy)$, then

$$re^{i\theta} = i \frac{\cos \pi y}{1 + \sin \pi y}.$$

This leads to two separate cases: either $0 < y \leq 1/2$ and $\theta = \pi/2$, or $1/2 \leq$

$y < 1$ and $\theta = -\pi/2$. In either case, show that

$$r^2 = \frac{1 - \sin \pi y}{1 + \sin \pi y} \quad \text{and} \quad P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}.$$

- (b) In the integral $\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi$ make the change of variables $t = F(e^{i\varphi})$. Observe that

$$e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}},$$

and then take the imaginary part and differentiate both sides to establish the two identities

$$\sin \varphi = \frac{1}{\cosh \pi t} \quad \text{and} \quad \frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}.$$

Hence deduce that

$$\begin{aligned} \frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi &= \frac{1}{2\pi} \int_0^\pi \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi} \tilde{f}_0(\varphi) d\varphi \\ &= \frac{\sin \pi y}{2} \int_{-\infty}^\infty \frac{f_0(t)}{\cosh \pi t - \cos \pi y} dt. \end{aligned}$$

- (c) Use a similar argument to prove the formula for the integral $\frac{1}{2\pi} \int_{-\pi}^0 P_r(\theta - \varphi) \tilde{f}_1(\varphi) d\varphi$.

8. Find a harmonic function u in the open first quadrant that extends continuously up to the boundary except at the points 0 and 1, and that takes on the following boundary values: $u(x, y) = 1$ on the half-lines $\{y = 0, x > 1\}$ and $\{x = 0, y > 0\}$, and $u(x, y) = 0$ on the segment $\{0 < x < 1, y = 0\}$.

[Hint: Find conformal maps $F_1, F_2, \dots, F_5$ indicated in Figure 11. Note that $\frac{1}{\pi} \arg(z)$ is harmonic on the upper half-plane, equals 0 on the positive real axis, and 1 on the negative real axis.]

9. Prove that the function u defined by

$$u(x, y) = \operatorname{Re} \left(\frac{i + z}{i - z} \right) \quad \text{and} \quad u(0, 1) = 0$$

is harmonic in the unit disc and vanishes on its boundary. Note that u is not bounded in $\mathbb{D}$.

10. Let $F : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function that satisfies

$$|F(z)| \leq 1 \quad \text{and} \quad F(i) = 0.$$

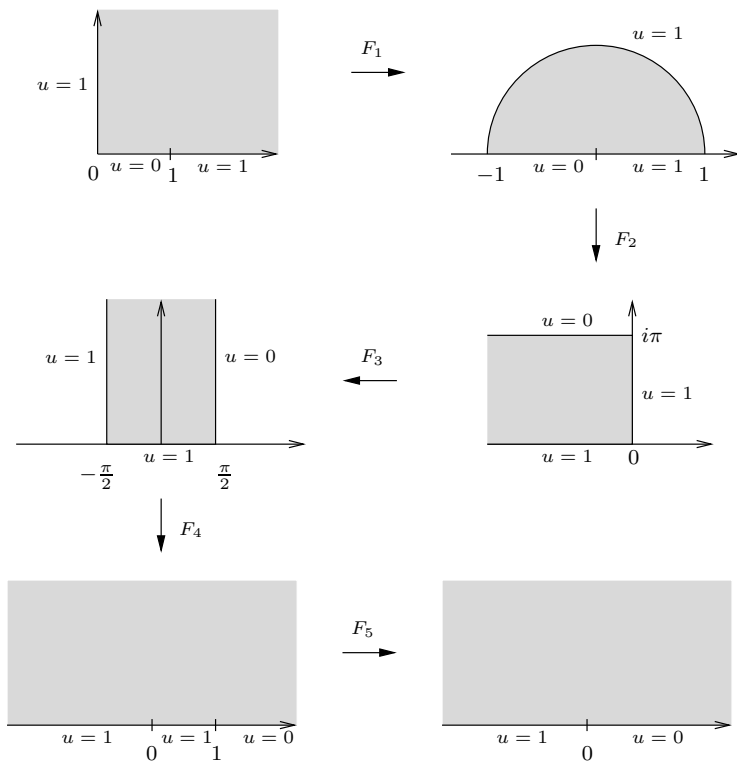


Figure 11. Successive conformal maps in Exercise 8

Prove that

$$|F(z)| \leq \left| \frac{z-i}{z+i} \right| \quad \text{for all } z \in \mathbb{H}.$$

11. Show that if $f : D(0, R) \rightarrow \mathbb{C}$ is holomorphic, with $|f(z)| \leq M$ for some $M > 0$, then

$$\left| \frac{f(z) - f(0)}{M^2 - \overline{f(0)}f(z)} \right| \leq \frac{|z|}{MR}.$$

[Hint: Use the Schwarz lemma.]

12. A complex number $w \in \mathbb{D}$ is a **fixed point** for the map $f : \mathbb{D} \rightarrow \mathbb{D}$ if $f(w) = w$.

- (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic and has two distinct fixed points, then f is the identity, that is, $f(z) = z$ for all $z \in \mathbb{D}$.

- (b) Must every holomorphic function $f : \mathbb{D} \rightarrow \mathbb{D}$ have a fixed point? [Hint: Consider the upper half-plane.]

13. The **pseudo-hyperbolic distance** between two points $z, w \in \mathbb{D}$ is defined by

$$\rho(z, w) = \left| \frac{z - w}{1 - \overline{w}z} \right|.$$

- (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, then

$$\rho(f(z), f(w)) \leq \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

Moreover, prove that if f is an automorphism of $\mathbb{D}$ then f preserves the pseudo-hyperbolic distance

$$\rho(f(z), f(w)) = \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

[Hint: Consider the automorphism $\psi_\alpha(z) = (z - \alpha)/(1 - \overline{\alpha}z)$ and apply the Schwarz lemma to $\psi_{f(w)} \circ f \circ \psi_w^{-1}$.]

- (b) Prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2} \quad \text{for all } z \in \mathbb{D}.$$

This result is called the Schwarz-Pick lemma. See Problem 3 for an important application of this lemma.

14. Prove that all conformal mappings from the upper half-plane $\mathbb{H}$ to the unit disc $\mathbb{D}$ take the form

$$e^{i\theta} \frac{z - \beta}{z - \overline{\beta}}, \quad \theta \in \mathbb{R} \text{ and } \beta \in \mathbb{H}.$$

15. Here are two properties enjoyed by automorphisms of the upper half-plane.

- (a) Suppose Φ is an automorphism of $\mathbb{H}$ that fixes three distinct points on the real axis. Then Φ is the identity.
- (b) Suppose (x_1, x_2, x_3) and (y_1, y_2, y_3) are two pairs of three distinct points on the real axis with

$$x_1 < x_2 < x_3 \quad \text{and} \quad y_1 < y_2 < y_3.$$

Prove that there exists (a unique) automorphism Φ of $\mathbb{H}$ so that $\Phi(x_j) = y_j$, $j = 1, 2, 3$. The same conclusion holds if $y_3 < y_1 < y_2$ or $y_2 < y_3 < y_1$.

16. Let

$$f(z) = \frac{i-z}{i+z} \quad \text{and} \quad f^{-1}(w) = i \frac{1-w}{1+w}.$$

- (a) Given $\theta \in \mathbb{R}$, find real numbers a, b, c, d such that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} \left(e^{i\theta} f(z) \right).$$

- (b) Given $\alpha \in \mathbb{D}$ find real numbers a, b, c, d so that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} (\psi_\alpha(f(z))),$$

with ψ_α defined in Section 2.1.

- (c) Prove that if g is an automorphism of the unit disc, then there exist real numbers a, b, c, d such that $ad - bc = 1$ and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} \circ g \circ f(z).$$

[Hint: Use parts (a) and (b).]

17. If $\psi_\alpha(z) = (\alpha - z)/(1 - \bar{\alpha}z)$ for $|\alpha| < 1$, prove that

$$\frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha|^2 dx dy = 1 \quad \text{and} \quad \frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha| dx dy = \frac{1 - |\alpha|^2}{|\alpha|^2} \log \frac{1}{1 - |\alpha|^2},$$

where in the case $\alpha = 0$ the expression on the right is understood as the limit as $|\alpha| \rightarrow 0$.

[Hint: The first integral can be evaluated without a calculation. For the second, use polar coordinates, and for each fixed r use contour integration to evaluate the integral in θ .]

18. Suppose that Ω is a simply connected domain that is bounded by a piecewise-smooth closed curve γ (in the terminology of Chapter 1). Then any conformal map F of $\mathbb{D}$ to Ω extends to a continuous bijection of $\mathbb{D}$ to $\bar{\Omega}$. The proof is simply a generalization of the argument used in Theorem 4.2.

19. Prove that the complex plane slit along the union of the rays $\cup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected.

[Hint: Given a curve, first “raise” it so that it is completely contained in the upper half-plane.]

20. Other examples of elliptic integrals providing conformal maps from the upper half-plane to rectangles are given below.

- (a) The function

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}, \quad \text{with } \lambda \in \mathbb{R} \text{ and } \lambda \neq 1$$

maps the upper half-plane conformally to a rectangle, one of whose vertices is the image of the point at infinity.

- (b) In the case
- $\lambda = -1$
- , the image of

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}}$$

is a square whose side lengths are $\frac{\Gamma^2(1/4)}{2\sqrt{2\pi}}$.

21. We consider conformal mappings to triangles.

- (a) Show that

$$\int_0^z z^{-\beta_1}(1-z)^{-\beta_2} dz,$$

with $0 < \beta_1 < 1$, $0 < \beta_2 < 1$, and $1 < \beta_1 + \beta_2 < 2$, maps $\mathbb{H}$ to a triangle whose vertices are the images of 0, 1, and ∞ , and with angles $\alpha_1\pi$, $\alpha_2\pi$, and $\alpha_3\pi$, where $\alpha_j + \beta_j = 1$ and $\beta_1 + \beta_2 + \beta_3 = 2$.

- (b) What happens when $\beta_1 + \beta_2 = 1$?
- (c) What happens when $0 < \beta_1 + \beta_2 < 1$?
- (d) In (a), the length of the side of the triangle opposite angle $\alpha_j\pi$ is $\frac{\sin(\alpha_j\pi)}{\pi}\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)$.

22. If P is a simply connected region bounded by a polygon with vertices $a_1, \dots, a_n$ and angles $\alpha_1\pi, \dots, \alpha_n\pi$, and F is a conformal map of the disc $\mathbb{D}$ to P , then there exist complex numbers $B_1, \dots, B_n$ on the unit circle, and constants c_1 and c_2 so that

$$F(z) = c_1 \int_1^z \frac{d\zeta}{(\zeta - B_1)^{\beta_1} \cdots (\zeta - B_n)^{\beta_n}} + c_2.$$

[Hint: This follows from the standard correspondence between $\mathbb{H}$ and $\mathbb{D}$ and an argument similar to that used in the proof of Theorem 4.7.]

23. If

$$F(z) = \int_1^z \frac{d\zeta}{(1-\zeta^n)^{2/n}},$$

then F maps the unit disc conformally onto the interior of a regular polygon with n sides and perimeter

$$2^{\frac{n-2}{n}} \int_0^\pi (\sin \theta)^{-2/n} d\theta.$$

24. The elliptic integrals K and K' defined for $0 < k < 1$ by

$$K(k) = \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \quad \text{and} \quad K'(k) = \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}}$$

satisfy various interesting identities. For instance:

(a) Show that if $\tilde{k}^2 = 1 - k^2$ and $0 < \tilde{k} < 1$, then

$$K'(k) = K(\tilde{k}).$$

[Hint: Change variables $x = (1 - \tilde{k}^2 y^2)^{-1/2}$ in the integral defining $K'(\tilde{k})$.]

(b) Prove that if $\tilde{k}^2 = 1 - k^2$, and $0 < \tilde{k} < 1$, then

$$K(k) = \frac{2}{1 + \tilde{k}} K\left(\frac{1 - \tilde{k}}{1 + \tilde{k}}\right).$$

[Hint: Change variables $x = 2t/(1 + \tilde{k} + (1 - \tilde{k})t^2)$.]

(c) Show that for $0 < k < 1$ one has

$$K(k) = \frac{\pi}{2} F(1/2, 1/2, 1; k^2),$$

where F the hypergeometric series. [Hint: This follows from the integral representation for F given in Exercise 9, Chapter 6.]

6 Problems

1. Let f be a complex-valued C^1 function defined in the neighborhood of a point z_0 . There are several notions closely related to conformality at z_0 . We say that f is **isogonal** at z_0 if whenever $\gamma(t)$ and $\eta(t)$ are two smooth curves with $\gamma(0) = \eta(0) = z_0$, that make an angle θ there ($|\theta| < \pi$), then $f(\gamma(t))$ and $f(\eta(t))$ make an angle of θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ . Also, f is said to be **isotropic** if it magnifies lengths by some factor for all directions emanating from z_0 , that is, if the limit

$$\lim_{r \rightarrow 0} \frac{|f(z_0 + re^{i\theta}) - f(z_0)|}{r}$$

exists, is non-zero, and independent of θ .